

JEE MAIN REVISION MASTER: REDOX REACTIONS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 21 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Oxidation Number Rules & Redox Titrations	21 PYQs	Redox Principles

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Oxidation Number Rules & Redox Titrations	$\text{Eq. Wt.} = \frac{M}{n}, \quad n = \text{factor of KMnO}_4: \begin{matrix} \text{Acidic} = 5, \\ \text{Neutral} = 3, \text{Strongly Basic} = 1 \end{matrix}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Oxidation Number Rules & Redox Titrations (Very High Yield — 21 PYQs)

Core Principle: Rules for oxidation states: Peroxides have O = -1, superoxides O = -1/2, in OF₂ oxygen is +2.

Disproportionation is a reaction where the same element is simultaneously oxidized and reduced (e.g. Cl₂ + NaOH).

Equivalent weight = (Molar Mass / n-factor). In acidic medium: KMnO₄ → Mn²⁺ (n=5); K₂Cr₂O₇ → 2Cr³⁺ (n=6).

■ **Caution / JEE Pitfall:** In Caro's acid (H₂SO₅) and Marshall's acid (H₂S₂O₈), sulphur has an oxidation state of +6, NOT +8 or +7, because of peroxide linkage (-O-O-)!

■ **Speed Trick:** Mnemonic for n-factors of KMnO₄: 'BAN 153' (Basic: 1, Acidic: 5, Neutral: 3).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): " $aC_2O_4^{2-} + bMnO_4^- + cH^+ \rightarrow xMn^{2+} + yH_2O + zCO_2$ a and ..." [Ans: A]
- [2026 Shift] Q#2 (MCQ): "Given below are two statements. Statement-I: In H₂O₂, each oxygen atom is assigned an oxidation number of -1, in ...". [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "In the titration of potassium permanganate (KMnO₄) against Ferrous ammonium sulphate (FAS) solution, dilute sulphuric acid but not ni..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

$a\text{C}_2\text{O}_4^{2-} + b\text{MnO}_4^- + c\text{H}^+ \rightarrow x\text{Mn}^{2+} + y\text{H}_2\text{O} + z\text{CO}_2$ a and x respectively are

- (A) 5,2 [Correct]
- (B) 4,1
- (C) 3,2
- (D) 4,2

Correct Answer: A

Solution Analysis: In acidic medium, MnO_4^- oxidises oxalate ion $\text{C}_2\text{O}_4^{2-}$ to CO_2 , and itself gets reduced to Mn^{2+} . First write the two half-reactions. For oxalate ion: $\text{C}_2\text{O}_4^{2-} \rightarrow 2\text{CO}_2$ Carbon is oxidised here, and the balanced half-reaction is: $\text{C}_2\text{O}_4^{2-} \rightarrow 2\text{CO}_2 + 2\text{e}^-$. For permanganate ion in acidic medium: $\text{MnO}_4^- \rightarrow \text{Mn}^{2+}$. Balancing oxygen and hydrogen using H_2O and H^+ : $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$